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Paper Code : ESCS201 Programming for Problem Solving

UPID : 002008

Time Allotted : 3 Hours

Full Marks : 70

*The Figures in the margin indicate full marks.**Candidate are required to give their answers in their own words as far as practicable***Group-A (Very Short Answer Type Question)**

1. Answer any ten of the following :

[1 x 10 = 10]

(i) What will be the output of the following C code?

```

1. #include <stdio.h>
2. void main()
3. {
4.   int a[2][3] = {1, 2, 3, 4, 5};
5.   int i = 0, j = 0;
6.   for (i = 0; i < 2; i++)
7.     for (j = 0; j < 3; j++)
8.       printf("%d", a[i][j]);
9. }

```

(ii) What is the worst case complexity of bubble sort?

(iii) What is the default return type of function definition?

(iv) What will be the output of the following C code?

```

#include<stdio.h>
main()
{
    int n;
    n=f1(4);
    printf("%d",n);
}
f1(int x)
{
    int b;
    if(x==1)
        return 1;
    else
        b=x*f1(x-1);
    return b;
}

```

(v) What is the size of a C structure?

(vi) What will be the output?

```

#include <stdio.h>
int main(){
    int a=3, *b = &a;
    printf("%d",a*b);
}

```

(vii) Which language is written in binary codes only?

(viii) The C-preprocessors are specified with which symbol?

(ix) What will be the output of the following C code? (Assuming that we have entered the value 1 in the standard input)

```

1. #include <stdio.h>
2. void main()
3. {
4.     double ch;
5.     printf("enter a value between 1 to 2:");
6.     scanf("%lf", &ch);
7.     switch (ch)
8.     {
9.         case 1:
10.            printf("1");
11.            break;
12.         case 2:
13.            printf("2");
14.            break;
15.     }
16. }

```

- (X) What will happen if in a C program you assign a value to an array element whose subscript exceeds the size of array?
- (X) What is the advantage of selection sort over other sorting techniques?
- (X) What is the limit for number of functions in a C Program?

Group-B (Short Answer Type Question)

Answer any three of the following :

[5 x 3 = 15]

2. Write the differences between call by value and call by reference. [5]
3. Write a program to compute factorial of a number using recursion. [5]
4. What is the difference between break and continue statement? [5]
5. Differentiate between Compiler and interpreter ? [5]
6. Write a C program to perform addition, subtraction, multiplication and division of two numbers by using switch case. [5]

Group-C (Long Answer Type Question)

Answer any three of the following :

[15 x 3 = 45]

7. Describe different types of asymptotic notation of complexity analysis of algorithms. [15]
8. (a) Write a C program to find the maximum and minimum of some values using a function that returns an array. [7]
- (b) Write a program in C to print all palindrome numbers in a given range using the function. [8]
9. (a) Write an algorithm to sort an array using the Merge sort algorithm [10]
- (b) Calculate the time complexity of Merge sort. [5]
10. What are the different types of Computer? [15]
11. (a) Write a program in C language to print the factorial of a number. [8]
- (b) Write a C program to print the Fibonacci series upto nth term. [7]